2026 NLP Homework 2

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Question 1

a.

Because $Y \subseteq Z$, $p(Z) = p(Y \cup \bar{Y}Z) = p(Y) + p(\bar{Y}Z)$, due to $\bar{Y}Z \cap Y = \emptyset$, and $p(\bar{Y}Z) \geq 0$, $p(Z) \geq p(Y)$

b.

$p(X \mid Z) = \frac{p(X \cap Z)}{p(Z)}$, because $(X \cap Z) \subseteq Z$, so $p(X \cap Z) \leq p(Z)$, so the probability $p(X \mid Z)$ in the range $[0, 1]$.

c.

Because $p(\mathcal{E} \cup \emptyset) = p(\mathcal{E}) + p(\emptyset) = p(\mathcal{E})$, so $p(\emptyset) = 0$

d.

$p(X \cup \bar{X}) = 1$, $p(X) + p(\mathcal{E} - X) = 1$, so $p(X) = 1 - p(\mathcal{E} - X) = 1 - p(\bar{X})$

e.

$p(s \cap r \mid r) = \frac{p(s \cap r \cap r)}{p(r)} = \frac{p(s \cap r)}{p(r)} = p(s \mid r)$

f.

$p(X \mid Y) = p(1 - \bar{X} \mid Y) = \frac{p((1-\bar{X}) \cap Y)}{p(Y)} = \frac{p(Y) - p(\bar{X} \cap Y)}{p(Y)} = 1 - p(\bar{X} \mid Y)$

g.

This equation is equal to:

$$
\begin{aligned}
&(p(XY) + p(X\bar{Y})) \cdot p(\bar{Z} \mid X)/p(\bar{Z}) \\
&= p(X) \cdot p(\bar{Z} \mid X)/p(\bar{Z}) \\
&= p(\bar{Z}X)/p(\bar{Z}) = p(X \mid \bar{Z})
\end{aligned}
$$

h.

According to the axiom, $p(singing \cup rainy) = p(singing) + p(rainy)$ when $singing \cap rainy = \emptyset$

i.

$p(singing \cap rainy) = p(singing \mid rainy) \cdot p(rainy)$. When singing and rainy are independent, which equals to singing $\cap$ rainy $= \emptyset$

j.

$p(X \mid Y) = \frac{p(XY)}{p(Y)} = 0$, so $p(XY) = 0$. Because $XY \cap Z \subseteq XY$, $p(XY \cap Z) \leq p(XY) = 0$, $p(X \mid Y, Z) = \frac{p(XY \cap Z)}{p(YZ)} = 0$

k.

Because $p(W \mid Y) = 1$, according to f, $1 - p(\bar{W} \mid Y) = 1$, so $p(\bar{W} \mid Y) = 0$. According to j, $p(\bar{W} \mid YZ) = 0$, $1 - p(W \mid YZ) = 0$, so $p(W \mid YZ) = 1$.

Question 2

a.

$p(Actual = blue \mid Claimed = blue) = \frac{p(Claimed=blue|Actual=blue) \cdot p(Actual=blue)}{p(Claimed=blue)}$

b.

hypothesis: Claimed = blue
evidence: Actual = blue
$p(Claimed = blue)$: prior probability of the hypothesis
$p(Claimed = blue \mid Actual = blue)$: likelihood of the hypothesis
$p(Actual = blue \mid Claimed = blue)$: posterior probability of the hypothesis

c.

p(Actual = blue) = 0.1
p(Claimed = blue | Actual = blue) = 0.8

$$
\begin{aligned}
&p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}) \\
&= \frac{p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) \cdot p(\text{Actual} = \text{blue})}{p(\text{Claimed} = \text{blue, Actual} = \text{blue}) + p(\text{Claimed} = \text{blue, Actual} = \text{red})} \\
&= \frac{p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) \cdot p(\text{Actual} = \text{blue})}{p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}) \cdot p(\text{Actual} = \text{blue}) \\
&\quad + p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{red}) \cdot p(\text{Actual} = \text{red})} \\
&= \frac{0.8 \cdot 0.1}{(0.8 \cdot 0.1) + (0.2 \cdot 0.9)} \\
&= \frac{0.08}{0.26} \approx 0.3077
\end{aligned}
$$

The judge should care more about $p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue})$, as this measures the true reliability of the witness's claim given the prior probabilities.

d.

$$
\begin{aligned}
p(A \mid B, Y) &= \frac{p(B, Y \mid A) \cdot p(A)}{p(B, Y)} \\
&= \frac{p(B, Y \mid A) \cdot p(A)}{p(B \mid Y) \cdot (Y)} \\
&= \frac{p(A, B, Y)}{p(B, Y)}
\end{aligned}
$$

$$
\begin{aligned}
\frac{p(B \mid A, Y) \cdot p(A \mid Y)}{p(B \mid Y)} &= \frac{p(B \mid A, Y) \cdot p(A, Y)}{p(B \mid Y) \cdot p(Y)} \\
&= \frac{p(B \mid A, Y) \cdot p(A, Y)}{p(B, Y)} \\
&= \frac{p(A, B, Y)}{p(B, Y)}
\end{aligned}
$$

e.

$$\begin{aligned} p(B \mid A, Y) \cdot p(A \mid Y) + p(B \mid \bar{A}, Y) \cdot p(\bar{A} \mid Y) &= \frac{p(B \mid A, Y) \cdot p(A, Y)}{p(Y)} + \frac{p(B \mid \bar{A}, Y) \cdot p(\bar{A}, Y)}{p(Y)} \\ &= \frac{p(A, B, Y)}{p(Y)} + \frac{p(\bar{A}, B, Y)}{p(Y)} \\ &= \frac{p(B, Y)}{p(Y)} \\ &= p(B \mid Y) \end{aligned}$$

f.

$$\begin{aligned} &p(\text{Actual} = \text{blue} \mid \text{Claimed} = \text{blue}, \text{city} = \text{Baltimore}) \\ &= \frac{p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}, \text{city} = \text{Baltimore}) \cdot p(\text{Actual} = \text{blue} \mid \text{city} = \text{Baltimore})}{p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{blue}, \text{city} = \text{Baltimore}) \cdot p(\text{Actual} = \text{blue} \mid \text{city} = \text{Baltimore}) \\ &\quad + p(\text{Claimed} = \text{blue} \mid \text{Actual} = \text{red}, \text{city} = \text{Baltimore}) \cdot p(\text{Actual} = \text{red} \mid \text{city} = \text{Baltimore})} \\ &= \frac{0.8 \cdot 0.1}{(0.8 \cdot 0.1) + (0.2 \cdot 0.9)} \\ &= \frac{0.08}{0.26} \approx 0.3077 \end{aligned}$$

Question 3

a.

$\sum_{cry} p(cry \mid situation) = 1$

b.

Table 1: Joint Probability Table $p(\text{cry}, \text{situation})$

$p(\text{cry}, \text{situation})$	**Predator!**	**Timber!**	**I need help!**	**TOTAL**
bwa	0.00	0.00	0.64	0.64
bwee	0.00	0.00	0.08	0.08
kiki	0.20	0.00	0.08	0.28
TOTAL	0.20	0.00	0.80	1.00

c.

i.

$$p(\text{Predator!} \mid \text{kiki})$$

ii.

$$\frac{p(\text{Predator!} \cap \text{kiki})}{p(\text{kiki})}$$

iii. Its value is:

$$\frac{0.20}{0.28} \approx 0.7143$$

iv.

$$\frac{p(\text{Predator!}) \cdot p(\text{kiki} \mid \text{Predator!})}{\begin{aligned} &p(\text{Predator!}) \cdot p(\text{kiki} \mid \text{Predator!}) \\ &+ p(\text{Timber!}) \cdot p(\text{kiki} \mid \text{Timber!}) \\ &+ p(\text{I need help!}) \cdot p(\text{kiki} \mid \text{I need help!}) \end{aligned}}$$

v. Using the above tables, the value of this is:

$$\frac{0.2 \cdot 1.0}{0.2 \cdot 1.0 + 0.0 \cdot 0.3 + 0.8 \cdot 0.1} = \frac{0.2}{0.28} \approx 0.7143$$

Question 4

a.

$p(w_1, w_2, w_3, w_4)$

$= p(w_1 \mid w_{-1}, w_0) \cdot p(w_2 \mid w_0, w_1) \cdot p(w_3 \mid w_1, w_2) \cdot p(w_4 \mid w_2, w_3) \cdot p(w_5 \mid w_3, w_4)$

$= p(w_1 \mid BOS, BOS) \cdot p(w_2 \mid BOS, w_1) \cdot p(w_3 \mid w_1, w_2) \cdot p(w_4 \mid w_2, w_3) \cdot p(EOS \mid w_3, w_4)$

$$= \frac{c(w_1, BOS, BOS)}{c(BOS, BOS)} \cdot \frac{c(w_2, BOS, w_1)}{c(BOS, w_1)} \cdot \frac{c(w_1, w_2, w_3)}{c(w_1, w_2)} \cdot \frac{c(w_2, w_3, w_4)}{c(w_2, w_3)} \cdot \frac{c(w_3, w_4, EOS)}{c(w_3, w_4)}$$

$c(BOS, BOS)$ is counted for the number of sentences in the corpus, and $c(BOS, BOS, i)$ means how many sentences started with i, and $c(bagpipe, music, EOS)$ means the number of sentences end with bagpipe music.

b.

I think because it is not a sentence, and it not finished yet, so the probability of having this sentence in the corpus is rare.
The $c(w_3, w_4, EOS)$ will be very low, because very little sentence ended with "think the".

c.

I think A is case 2, because it neither contains BOS, nor contains EOS, and B is 1 because the BOS indicates it starts with do and EOS indicates it ends with you think. And C is case 3 because it just has BOS and has no EOS indicating ending.

Question 5

$\frac{p(a)}{p(b)} = \frac{p(their|think) \cdot p(okay|their)}{p(they're|think) \cdot p(okay|they're)}$

We should count 4 parameters to compute. If it uses 5-gram model, we should count 10 parameters because we should count word "their", "okay", "by", "now" and "EOS" 's conditional probabilities (because the condition has their), same for "they're".

Question 6

We could define a prior probability for each topic say p(k), and when we compute the bigram model, say $p(w_2)$, we can count it as $p(w_2 \mid w_1, k) \cdot p(k)$, and sum over the total $\sum_k p(k)$ to get the $p(w_2)$. So in case of $p(w_1 w_2 w_3 w_4)$, we could refer it to $\sum_k p(w_1 w_2 w_3 w_4 \mid k) = \sum_k p(k) \cdot p(w_1) \cdot p(w_2 \mid w_1, k) \cdot p(w_3 \mid w_2, k) \cdot p(w_4 \mid w_3, k)$

Question 9

a.

The 10 words most similar to seattle are:
seahawks dallas atlanta wichita tacoma lauderdale florida spokane chino dulles
dog:
dogs badger cat hound puppy dachshund sighthound poodle rat keeshond
communist:
socialist bolshevik communists comintern trotskyist leftist bolsheviks cominform stalinist leninist
jpg:
png svg szczepanek buteo pix gif image galleria regnum fiav
the:
in its of which entire within from a part second
google:
webpage yahoo web search com blogging faq editable archived redirection

For seattle, seahawks is famous in Seattle, that makes sense, and the behind words are all city name, which are fine. And for dog, most words are a special type of a dog, this is also good. For jpg, the words szczepanek, buteo, fiav are

wired, I do not know what does it mean. Also galleria, regnum seem to have no relationship to jpg. And png, svg, pix, gif, image are pretty good, those are all related to the type of an image. For the and google, those similar 10 words make sense.

I think the most similar words are all related to a type, for example, the is det, so most similar words are prep or det.

The less features embedded, the less concrete the information is. For example, when I use words-10, seattle, it gives me indianapolis atlanta lakers dallas expos boston detroit cleveland houston texas, it is more broad and has more "variance"", while words-200 it gives me tacoma spokane seahawks redskins intelligencer bremerton dulles denzel dc yamasaki, those most 3 similar words are all cities in seattle, which are more specific and is more similar to the word itself, making the most of the word feature.

b.

The most similar 10 words to king - man + woman are: queen marries isabella sibylla consort anjou vasa heiress valois betrothed, and car - road + air gives me: helicopter aircraft pressurized airframe psa interceptor usaf curtiss usmc parachutes, those two work pretty well. I think hitler - germay + italy words not so well, it should generate some words about the leaders in italy playing role as hitler in Germany, the first word mussolini and himmler is fine, but the following three words: rome loos speer do not mean that, maybe because of the less of that in italy.

When it comes to word-10, it does not work well, I think because the 10 embeddings catch information of the word much less than the word-100, and when it comes to add or minus operation, those meaning even fades a lot. king - man means no male kings, which is the result of subtracting the man attribute in the king, and then we plus woman, add the female attribute, thus leading to the word queen as the most similar word to this operation. And car - road + air is the same case, when car - road, it indicates the vehicle that does not drive on the road but in the air, which leads to the helicopter. So when we do the analogies, we could do some operations between high dimension vectors of words, and by subtracting or adding the attributes we could obtain the new terminology.